Fourier Optics and Structured Light

Band-limited angular-spectrum propagation, Gaussian and Laguerre–Gauss beams, the Talbot carpet, and where the method fails

Kevin

Abstract

A monochromatic field in a plane is a superposition of plane waves, and free propagation multiplies each by $e^{ik_z z}$. We implement this angular-spectrum propagator with the Matsushima–Shimobaba band limit, optional zero padding and a paraxial switch, and check it against plane waves, Parseval (10^{-15}) and an exact Hankel integral (10^{-9}). Applied to Gaussian beams, Laguerre–Gauss (LG) vortex modes and Talbot self-imaging, every deviation from the paraxial textbook formulas is shown to be physics, not numerical error, and is derived in closed form with no fitted parameter: the nonparaxial width $w_0^2[1+\zeta^2(1+4\varepsilon+24\varepsilon^2)]$ (matched to 2×10^{-11}), the Gouy-phase correction $2\varepsilon\zeta(1-\zeta^2)/(1+\zeta^2)^2$ (10^{-8} rad), and the LG overlap phase and shape loss (to 1%). Vortex modes have integer winding and OAM $l\hbar$ per photon to 10^{-12}, and a two-mode superposition rotates rigidly by the Gouy phase. For the Talbot effect we show that only a cosine grating self-images exactly, that the paraxial Talbot length is always $\lambda/2$ too long, and that a binary grating's defect obeys $1-F\simeq 0.5137\sqrt{\lambda/d}$, derived via a Gamma-function integral and confirmed to four digits at $d/\lambda=10^5$. Finally we map the sampling limits of the method: critical distance, window size and grid spacing.

1 Introduction

Scalar diffraction theory [1] underlies microscope design, holography, beam shaping and optical trapping. Its numerical workhorse is the angular-spectrum method: Fourier transform the field, multiply by the free-space transfer function, transform back. The method is exact for the scalar Helmholtz equation, yet it has sampling pitfalls [3, 6], and the analytic results it is usually checked against (Gaussian beam width, Gouy phase, Talbot distance) are themselves paraxial approximations.

This project therefore has two goals: (i) build a propagator whose numerical error is understood and bounded, and (ii) use it to resolve the differences between exact and paraxial optics into closed-form, checkable corrections. Along the way it covers structured light, i.e. optical vortices carrying orbital angular momentum [4], and the Talbot carpet [5].

Contributions.

1. A band-limited angular-spectrum propagator with seven independent correctness checks and a mapped region of validity.
2. Closed-form nonparaxial corrections for the Gaussian width and Gouy phase and for the LG overlap phase and fidelity, matched with no fit.
3. Integer winding and OAM of LG modes and a visible Gouy rotation, with negative controls.
4. Talbot: exact versus paraxial self-imaging, the half-Talbot shift, and a derived $\sqrt{\lambda/d}$ defect law for binary gratings.

Table 1: Summary of the main results. $\lambda = 0.5\,\mu$m, $w_0 = 10\,\mu$m, $z_R = 628.3\,\mu$m, $\varepsilon = 1/(kw_0)^2 = 6.33 \times 10^{-5}$ unless stated.

Quantity	Computed	Analytic
Power after $z = 10\,\mu\text{m} \ldots 10$ cm	$\lvert P/P_0 - 1\rvert \le 10^{-15}$	conserved
Plane wave at 70° after 123 μm	$\le 2 \times 10^{-14}$	$E_0 e^{ik_z z}$
On-axis field vs exact Hankel integral, $\lvert z\rvert \le 3z_R$	$\le 10^{-9}$	exact
Gaussian width vs paraxial / vs nonparaxial formula	1.14×10^{-4} / $\le 2 \times 10^{-11}$	Eq. (19)
Gouy phase vs $\arctan\zeta$ / vs corrected	3.2×10^{-5} / $\le 10^{-8}$ rad	Eq. (24)
LG Gouy residual vs $-\zeta\varepsilon\langle s^2\rangle/4$	$\le 2.5 \times 10^{-6}$ rad	Eq. (30)
LG shape loss $1 - F$ at $3z_R$	$1.9 \times 10^{-7} \ldots 1.2 \times 10^{-5}$, within 1%	$(\zeta\varepsilon/4)^2 \mathrm{Var}(s^2)$
Phase winding, OAM per photon	l to 10^{-12}	l
Lobe rotation of LG_0^0+LG_0^2	$\arctan\zeta$ to 2×10^{-16} (paraxial TF)	$\arctan\zeta$
Cosine-grating revival, $d = 10\,\mu$m	399.749846 μm	$z_T = 399.74984$; paraxial 400
Half-Talbot shift of first intensity harmonic	$0.500000000000\,d$	$d/2$
Binary grating $F(z_T)$, $d = 10\,\mu$m	0.866	1 (paraxial)
Binary-grating defect, $d/\lambda \to \infty$	$0.5137\sqrt{\lambda/d}$, ratio 1.0000 at 10^5	Eq. (38)
Slit at 4 mm: no band limit / band-limited (padded)	6.9% / 0.05% error	—

2 Theory and derivations

Conventions. Time dependence $e^{-i\omega t}$, so a forward wave is e^{ikz}; $k = 2\pi n/\lambda$. Spatial frequencies f are in cycles per length, $k_x = 2\pi f_x$, and $k_\rho^2 = k_x^2 + k_y^2$. For a beam of waist w_0: $z_R = kw_0^2/2$, $\zeta = z/z_R$, $\psi = \arctan\zeta$ and $\varepsilon = 1/(kw_0)^2$.

2.1 Angular spectrum

Each Cartesian component of a monochromatic field in a homogeneous medium obeys the Helmholtz equation,

$$(\nabla^2 + k^2)E = 0. \tag{1}$$

Fourier transforming in the transverse coordinates, $\tilde{E}(k_x, k_y; z) = \iint E\, e^{-i(k_x x + k_y y)} dx\, dy$, replaces $\partial_x^2 + \partial_y^2$ by $-k_\rho^2$:

$$\partial_z^2 \tilde{E} + k_z^2 \tilde{E} = 0, \qquad k_z = \sqrt{k^2 - k_\rho^2}. \tag{2}$$

The solutions are $e^{\pm ik_z z}$. Keeping only the forward wave,

$$\tilde{E}(k_x, k_y; z) = \tilde{E}(k_x, k_y; 0)\; H(k_x, k_y; z), \qquad H = e^{ik_z z}. \tag{3}$$

This is exact for the scalar Helmholtz equation: each plane-wave component propagates independently, and the transfer function H is a pure phase for $k_\rho < k$.

Evanescent waves. For $k_\rho > k$, $k_z = i\kappa$ with $\kappa = \sqrt{k_\rho^2 - k^2}$, and $H = e^{-\kappa z}$ decays. For back-propagation ($z < 0$) we use $e^{-\kappa|z|}$, which keeps the operator bounded.

Paraxial limit. Expanding $k_z = k\sqrt{1-q}$ in $q = k_\rho^2/k^2$,

$$k_z = k - \frac{k_\rho^2}{2k} - \frac{k_\rho^4}{8k^3} - \frac{k_\rho^6}{16k^5} - \ldots \tag{4}$$

The first two terms give the Fresnel transfer function $H_{\text{par}} = e^{ikz} e^{-ik_\rho^2 z/2k}$, which is the exact propagator of the paraxial wave equation

$$2ik\,\partial_z A + \nabla_\perp^2 A = 0, \qquad E = A\, e^{ikz}. \tag{5}$$

Every nonparaxial correction derived below comes from the quartic term of Eq. (4).

Energy conservation. By Parseval's theorem the power is $P = \int |\tilde{E}|^2 d^2k/(2\pi)^2$. $|H| = 1$ for propagating components, so P is conserved exactly by Eq. (3); only the evanescent part decays.

2.2 Impulse response: Fresnel and Fraunhofer diffraction

Multiplication by H in the frequency domain is a convolution in real space,

$$E(\mathbf{r}_\perp, z) = \iint h(\mathbf{r}_\perp - \mathbf{r}'_\perp, z)\, E(\mathbf{r}'_\perp, 0)\, d^2 r'_\perp, \qquad h = \mathcal{F}^{-1}[H]. \tag{6}$$

For the exact H, h is the Rayleigh–Sommerfeld kernel. For the paraxial H_{par} the inverse transform is a Gaussian integral and gives the Fresnel kernel

$$h_F(\mathbf{r}_\perp, z) = \frac{e^{ikz}}{i\lambda z} \exp\Big(\frac{ik|\mathbf{r}_\perp|^2}{2z}\Big). \tag{7}$$

Expanding $|\mathbf{r}_\perp - \mathbf{r}'_\perp|^2 = r_\perp^2 - 2\mathbf{r}_\perp \cdot \mathbf{r}'_\perp + r'^2_\perp$ and dropping the last term, which is allowed when the source of size a has a small Fresnel number $a^2/\lambda z \ll 1$, gives the Fraunhofer limit

$$E(\mathbf{r}_\perp, z) \approx \frac{e^{ikz} e^{ikr_\perp^2/2z}}{i\lambda z}\, \tilde{E}\Big(\mathbf{k}_\perp = \frac{k\mathbf{r}_\perp}{z}\Big), \tag{8}$$

in which the far field is the angular spectrum itself, evaluated at the direction $\mathbf{r}_\perp/z$. The angular-spectrum method contains both limits; it never evaluates the kernel, which is highly oscillatory and hard to sample at short distances.

2.3 The discrete Fourier transform

On a grid of N points with spacing Δx and window $L = N\Delta x$, the discrete Fourier transform samples a spectrum at $f_j = j/L$. Two approximations are made at once:

- *Sampling in x* makes the spectrum periodic with period $1/\Delta x$. Spectral content beyond $|f| = 1/2\Delta x$ is folded back (aliased), so the field must be band-limited to the Nyquist frequency.
- *Truncation to the window* makes the field periodic with period L. Multiplying the DFT by H therefore implements a *circular* convolution: light that leaves the window on one side re-enters on the other.

Within these two conditions the discrete propagation is exact for the sampled field; there is no stepping error in z. The band limit and padding below address the second condition, and the grid study addresses the first.

2.4 Sampling: band limit and critical distance

On a numerical window of size L with N points of spacing Δx, the transfer function is sampled at frequency spacing $\Delta f = 1/L$. Its phase $\phi(f) = 2\pi z\sqrt{1/\lambda^2 - f^2}$ has local frequency (in the conjugate variable, a length)

$$\frac{1}{2\pi}\Big|\frac{d\phi}{df}\Big| = \frac{z\lambda f}{\sqrt{1 - \lambda^2 f^2}}. \tag{9}$$

The sampling theorem in f requires this to be at most $L/2$. Solving for f gives the band limit

$$|f| \le f_{\text{lim}} = \frac{1}{\lambda\sqrt{(2z/L)^2 + 1}}. \tag{10}$$

The limit only removes components that the grid can represent, $|f| < 1/(2\Delta x)$, when $f_{\text{lim}} < 1/(2\Delta x)$, i.e. beyond the critical distance

$$z_c = \frac{L}{2}\sqrt{\Big(\frac{2\Delta x}{\lambda}\Big)^2 - 1}. \tag{11}$$

Physical meaning. A plane wave of frequency f travels at angle θ with $\sin\theta = \lambda f$, and walks sideways by $z\tan\theta$. If this exceeds $L/2$, the light leaves the window and, because the discrete Fourier transform is periodic, re-enters from the other side. This wrap-around *is* the aliasing; removing those components is the correct answer inside a finite window.

Periodic fields. For a field that is itself periodic with period L (one grating period), the DFT is the exact Fourier series and wrap-around is physical. The band limit must then be switched off: at the Talbot distance it would give $f_{\lim} = 0.025\,\mu\mathrm{m}^{-1} < 1/d$ and delete every diffraction order.

No `fftshift`. H depends only on f, and the forward and inverse transforms use the same frequency grid, so where $x = 0$ sits in the array does not matter. A shift is only needed to plot a spectrum; building H on a shifted grid and multiplying it into an unshifted spectrum is wrong by $O(1)$.

2.5 Gaussian beam in the paraxial approximation

Try a solution of the paraxial equation of the form

$$A(r,z) = \frac{q_0}{q(z)} \exp\Big(\frac{ikr^2}{2q(z)}\Big). \tag{12}$$

Substituting and collecting powers of r, the r^2 terms require $q'(z) = 1$ and the r^0 terms then fix the prefactor $1/q$. So $q(z) = z - iz_R$, and

$$\frac{1}{q} = \frac{z}{z^2+z_R^2} + i\,\frac{z_R}{z^2+z_R^2} \equiv \frac{1}{R(z)} + \frac{2i}{kw^2(z)}. \tag{13}$$

Reading off the real and imaginary parts,

$$w^2(z) = w_0^2(1+\zeta^2), \qquad R(z) = z\big(1+\zeta^{-2}\big), \qquad w_0^2 = \frac{2z_R}{k}. \tag{14}$$

The prefactor is $q_0/q = -iz_R/(z-iz_R) = 1/(1+i\zeta) = e^{-i\psi}/\sqrt{1+\zeta^2}$. The full beam is therefore

$$E = \frac{w_0}{w}\, e^{-r^2/w^2}\, \exp\Big[i\Big(kz + \frac{kr^2}{2R} - \psi\Big)\Big], \qquad \psi = \arctan\zeta. \tag{15}$$

The phase $-\psi$ is the Gouy phase: on axis the beam *lags* a plane wave by $\arctan\zeta$, a total of π through the focus.

2.6 Gaussian beam beyond the paraxial approximation

Width, exactly. In k-space, x acts as $i\partial_{k_x}$. Applying this to $\tilde{E}\,e^{ik_z z}$ gives the operator identity

$$x(z) = x(0) + z\,\frac{k_x}{k_z}, \tag{16}$$

the ray equation for each plane-wave component. Squaring and averaging, the cross term $\langle x\,k_x/k_z + k_x/k_z\,x\rangle$ vanishes for a field that is real at $z = 0$, so

$$\langle x^2\rangle_z = \langle x^2\rangle_0 + z^2\Big\langle\frac{k_x^2}{k_z^2}\Big\rangle, \tag{17}$$

exactly. Now expand $k_x^2/k_z^2 = (k_x^2/k^2)(1+q+q^2+\dots)$ and use the Gaussian moments of $|\tilde{E}|^2 \propto e^{-k_\rho^2 w_0^2/2}$, for which each component has variance $\sigma^2 = 1/w_0^2$:

$$\langle k_x^2\rangle = \sigma^2, \qquad \langle k_x^2 k_\rho^2\rangle = 4\sigma^4, \qquad \langle k_x^2 k_\rho^4\rangle = 24\sigma^6. \tag{18}$$

With $\langle x^2\rangle_0 = w_0^2/4$ and $w^2 = 4\langle x^2\rangle$,

$$\frac{w^2(z)}{w_0^2} = 1 + \zeta^2\big(1 + 4\varepsilon + 24\varepsilon^2 + O(\varepsilon^3)\big). \tag{19}$$

The paraxial result is the leading term; the beam spreads slightly faster than paraxial theory predicts.

On-axis phase. With $s = k_\rho w_0/2$, the on-axis field is a Hankel integral,

$$\frac{E(0,z)}{E(0,0)} = 2\int_0^\infty e^{-s^2}\, e^{ik_z z}\, s\, ds, \tag{20}$$

which is the exact Helmholtz answer. From Eq. (4),

$$k_z z = kz - \zeta s^2 - \varepsilon\zeta s^4 + \dots \tag{21}$$

Expanding $e^{-i\varepsilon\zeta s^4} \approx 1 - i\varepsilon\zeta s^4$, and using, with $a = 1 + i\zeta$,

$$\int_0^\infty 2s\, e^{-as^2} ds = \frac{1}{a}, \qquad \int_0^\infty 2s^5 e^{-as^2} ds = \frac{2}{a^3}, \tag{22}$$

gives

$$\frac{E(0,z)}{E(0,0)} = \frac{e^{ikz}}{a}\Big(1 - \frac{2i\varepsilon\zeta}{a^2}\Big). \tag{23}$$

Taking the argument, with $\arg a = \psi$ and $\mathrm{Im}(-2i\varepsilon\zeta/a^2) = -2\varepsilon\zeta(1-\zeta^2)/(1+\zeta^2)^2$,

$$kz - \arg E(0,z) = \arctan\zeta + 2\varepsilon\,\frac{\zeta(1-\zeta^2)}{(1+\zeta^2)^2} + O(\varepsilon^2). \tag{24}$$

The correction vanishes at the waist and at $z = z_R$, and changes sign there.

2.7 Laguerre–Gauss modes and orbital angular momentum

In cylindrical coordinates the paraxial equation separates, and its solutions with azimuthal dependence $e^{il\phi}$ are the LG modes

$$\mathrm{LG}_{pl} = \frac{C_{pl}}{w}\, s^{|l|/2} L_p^{|l|}(s)\, e^{-r^2/w^2} e^{il\phi}\; e^{ikz + ikr^2/2R}\; e^{-i(2p+|l|+1)\psi}, \tag{25}$$

with $s = 2r^2/w^2$, $L_p^{|l|}$ a generalised Laguerre polynomial and $C_{pl} = \sqrt{2\,p!/[\pi(p+|l|)!]}$. They are orthonormal in each plane, and the mode order $2p + |l|$ multiplies the Gouy phase.

Gouy phase of a vortex. LG_{pl} vanishes on axis for $l \neq 0$, so its Gouy phase cannot be read on axis. Instead it is read from the overlap with the analytic mode without its Gouy factor:

$$\arg\big\langle \mathrm{LG}_{pl}\big|_{\psi=0} \big| E(z)\big\rangle = -(2p + |l| + 1)\psi. \tag{26}$$

Phase winding. Around any closed loop enclosing the axis, $\arg E$ increases by $2\pi l$. The winding number

$$W = \frac{1}{2\pi}\oint d(\arg E) \tag{27}$$

is an integer, and changes only when a zero of E crosses the loop. It is therefore topologically protected under propagation.

Orbital angular momentum. The OAM operator is $L_z = -i\hbar\,\partial_\phi = -i\hbar(x\partial_y - y\partial_x)$. For $e^{il\phi}$ it gives $l\hbar$ per photon. L_z commutes with k_ρ^2, hence with H, so OAM is conserved by both exact and paraxial propagation. For a superposition $\sum_l c_l\,\mathrm{LG}_{0l}$ the OAM per photon is the weighted mean $\sum_l |c_l|^2 l$, which need not be an integer, while the winding around the axis is still an integer.

Gouy rotation. For $(\mathrm{LG}_{00}+\mathrm{LG}_{02})/\sqrt{2}$, the two modes have Gouy orders 1 and 3. The intensity cross term is then $\propto \cos(2\phi - 2\psi)$, so the two-lobed pattern rotates rigidly by $\psi(z)$: 90° from the waist to the far field. The Gouy phase, invisible for a single mode, becomes a visible rotation.

Nonparaxial overlap. All propagators are functions of k_ρ and commute, so

$$U_{\mathrm{exact}} = U_{\mathrm{par}}\, e^{-izQ}, \qquad Q = \frac{k_\rho^4}{8k^3} + \dots \tag{28}$$

The overlap of the paraxially and exactly propagated mode is therefore

$$\langle U_{\mathrm{par}}\mathrm{LG}|U_{\mathrm{exact}}\mathrm{LG}\rangle = \langle \mathrm{LG}|e^{-izQ}|\mathrm{LG}\rangle. \tag{29}$$

The Fourier transform of LG_{pl} at the waist is again LG_{pl}, with $s = k_\rho^2 w_0^2/2$ in k-space, so $zQ = \zeta\varepsilon s^2/4$. Expanding in cumulants, $\langle e^{-iX}\rangle \approx \exp\!\big(-i\langle X\rangle - \tfrac{1}{2}\mathrm{Var}X\big)$, gives the phase and the fidelity:

$$\begin{aligned} \delta\arg &= -\frac{\zeta\varepsilon}{4}\,\langle s^2\rangle, \\ 1 - F &= \Big(\frac{\zeta\varepsilon}{4}\Big)^2 \mathrm{Var}(s^2), \\ \langle s^2\rangle &= 6p^2 + 6p|l| + l^2 + 6p + 3|l| + 2. \end{aligned} \tag{30}$$

For the Gaussian this gives $-\zeta\varepsilon/2$, not Eq. (24). The two are different quantities: the quartic term also couples the Gaussian to $p > 0$ modes, which are nonzero on axis, so the on-axis phase is not the overlap phase.

2.8 Talbot effect

A field periodic in x with period d contains only the frequencies $f_m = m/d$:

$$E(x,0) = \sum_m c_m\, e^{2\pi i m x/d}, \qquad E(x,z) = \sum_m c_m\, e^{2\pi i m x/d}\, e^{ik_{z,m}z}. \tag{31}$$

Exact revival distance. Harmonic m rephases with $m = 0$ when $(k - k_{z,m})z = 2\pi$. With $u = m^2\lambda^2/d^2$,

$$z_m = \frac{\lambda}{1 - \sqrt{1-u}} = \frac{\lambda\big(1 + \sqrt{1-u}\big)}{u}. \tag{32}$$

Expanding for $m = 1$,

$$z_T = \frac{2d^2}{\lambda} - \frac{\lambda}{2} + O\Big(\frac{\lambda^3}{d^2}\Big). \tag{33}$$

The paraxial Talbot length $2d^2/\lambda$ is too long by $\lambda/2$ for any period.

Only a cosine grating self-images. At z_T the phase of harmonic m relative to $m = 0$ is

$$(k - k_{z,m})\, z_T = 2\pi m^2\Big[1 + (m^2-1)\frac{\lambda^2}{4d^2} + \dots\Big]. \tag{34}$$

Only $m = 0, \pm 1$ rephase together exactly. A cosine grating, which has only these orders, self-images perfectly; any grating with higher orders does not. In the paraxial approximation every phase is exactly $2\pi m^2$ at $2d^2/\lambda$, which hides the effect.

Half-Talbot shift. At $z_T/2$ the first harmonic has phase π, so the first harmonic of the intensity is shifted by exactly $d/2$ for every grating, even though the full pattern is not a shifted copy.

Binary-grating defect. For a 50% binary grating, $|c_0|^2 = \frac{1}{4}$ and $|c_m|^2 = 1/(\pi m)^2$ for odd m. With $a = \lambda/d$, the phase error of order m at z_T is $\delta_m \simeq \frac{\pi}{2} m^4 a^2$. The overlap with the initial field is

$$S = P - \frac{2}{\pi^2} \sum_{m \text{ odd}} \frac{1 - e^{i\delta_m}}{m^2}. \tag{35}$$

The summand varies on the scale $m \sim a^{-1/2} \gg 1$, so the sum can be converted to an integral with $v = ma^{1/2}$ (the factor $\frac{1}{2}$ for odd m cancels the two signs of m):

$$1 - F \simeq \frac{4}{\pi^2} \operatorname{Re} J \sqrt{\frac{\lambda}{d}}, \qquad J = \int_0^\infty \frac{1 - e^{i\pi v^4/2}}{v^2}\, dv. \tag{36}$$

Substituting $s = (\pi/2)v^4$ turns $\operatorname{Re} J$ into a standard integral,

$$\int_0^\infty \frac{1 - \cos s}{s^{1+\alpha}}\, ds = -\Gamma(-\alpha) \cos\frac{\pi\alpha}{2}, \qquad \alpha = \tfrac{1}{4}, \tag{37}$$

and so

$$\operatorname{Re} J = \left(\frac{\pi}{2}\right)^{1/4} \frac{-\Gamma(-\frac{1}{4})}{4} \cos\frac{\pi}{8} = 1.26745, \qquad 1 - F \simeq 0.51368\sqrt{\lambda/d}. \tag{38}$$

The slow square-root approach to perfect imaging comes from the $1/m^2$ power tail of the spectrum of a sharp edge.

3 Numerical methods

3.1 Angular-spectrum propagator

The propagator implements Eq. (3) on a uniform grid:

1. Optionally zero-pad the field to twice the window, to stop wrap-around.
2. Take the discrete Fourier transform (FFT), which samples $\tilde{E}$ at $f = m/L$.
3. Multiply by $H(f; z)$, either exact ($e^{ik_z z}$, with $e^{-\kappa|z|}$ for evanescent components) or paraxial ($e^{ikz - i\pi\lambda z f^2}$), and optionally by the band-limit mask $\Theta(f_{\lim} - |f_x|)\,\Theta(f_{\lim} - |f_y|)$.
4. Inverse FFT, and crop if padded.

The cost is $O(N \log N)$ per step and independent of z; there is no stepping error, since each step is exact for the sampled spectrum. The same code handles one- and two-dimensional fields.

3.2 Beam diagnostics

- *Width:* $w = 2\sqrt{\langle x^2 \rangle}$ from the second moment of $|E|^2$, which is the quantity Eq. (19) predicts (not a fit, not a FWHM).
- *Curvature:* a weighted least-squares fit of $a + kx^2/2R$ to the unwrapped phase over $|x| < w$.
- *Gouy phase:* $kz - \arg E(0, z)$ for the Gaussian; for LG modes, the phase of the overlap integral with the analytic mode without its Gouy factor, evaluated by summation over the grid.
- *Fidelity:* $F = |\langle E_1 | E_2 \rangle|^2 / (\langle E_1 | E_1 \rangle \langle E_2 | E_2 \rangle)$.
- *Winding:* the sum of wrapped phase differences around a closed square loop of pixels, divided by 2π. Each difference is the principal value, so the result is an integer by construction.
- *OAM:* $\langle E | -i(x\partial_y - y\partial_x) | E \rangle / \langle E | E \rangle$, with the derivatives computed spectrally (multiplication by ik_x, ik_y in Fourier space), which is exact for band-limited fields.
- *Lobe orientation:* $\frac{1}{2} \arg \sum I\,(x + iy)^2$, a smooth weight that avoids the grid artefact of sampling $e^{2i\phi}$ directly.

3.3 Exact references

The on-axis Hankel integral is evaluated by 2000-node Gauss–Legendre quadrature on $s \in [0, 7]$. The integrand is bounded by $s\,e^{-s^2}$, so the neglected tail is below e^{-49}, and the quadrature does not depend on the propagation grid. The moments $\langle s^2 \rangle$ and $\mathrm{Var}(s^2)$ of the LG modes are integrals of a polynomial times e^{-s}, which Gauss–Laguerre quadrature evaluates exactly once the number of nodes exceeds half the polynomial degree.

3.4 Talbot computations

One grating period is used as a periodic window with $M = 512$ points and the band limit off, so the DFT is the exact Fourier series of the sampled grating. The revival peak is located by sampling $F(z)$ at 11 points spaced $0.002 z_T$ around the *paraxial* guess and fitting a parabola through the top three, so the method is not told the answer. With samples F_{i-1}, F_i, F_{i+1} around the largest value and spacing Δz, the vertex of the parabola is at

$$z^* = z_i + \frac{\Delta z}{2} \frac{F_{i-1} - F_{i+1}}{F_{i-1} - 2F_i + F_{i+1}}. \tag{39}$$

For large d/λ the binary-grating defect is computed from a closed-form phase sum over the propagating orders, which matches the FFT propagator to 4×10^{-13} where both apply.

3.5 Sampling study

A $20\,\mu$m slit on a 1024-point grid ($\Delta x = \lambda$, $L = 512\,\mu$m) is propagated with and without band limit and padding, and compared with the same propagation on a 64 times wider window, which is free of wrap-around at all distances considered. The window study varies the half-window relative to w at $3z_R$; the grid study varies $\Delta x / w_0$ and compares with an over-resolved reference.

4 Verification

The checks fall into four groups: exact properties of the propagator, analytic beam results, topology, and the limits of the method.

1. Exact properties of the propagator.

- *Unitarity:* power is conserved inside the passband to 10^{-15} up to $z = 10^5\,\mu$m; for a field with evanescent components, the evanescent power decays and the propagating power is preserved.
- *Eigenfunctions:* a plane wave at up to 70° (also in $n = 1.5$) stays a plane wave with phase $e^{ik_z z}$. This tests the transfer function pointwise.
- *Group property:* propagating forward then backward is the identity, and $H(z_1)H(z_2) = H(z_1 + z_2)$.
- *Symmetry:* the result commutes with translations without any `fftshift`; as a negative control, building H on a shifted grid gives a phase ramp and an $O(1)$ error. A one-dimensional field equals a y-invariant slice of the two-dimensional propagation.
- *Approximation:* the paraxial transfer function agrees with the exact one at small angles within the size of the quartic term.

2. Analytic beam results. Each closed form of Section 2 is compared with the propagated field: Gaussian width, curvature and Gouy phase, their nonparaxial corrections, and the on-axis field against the exact Hankel integral. For LG modes: orthonormality; the closed form satisfies the paraxial equation to 10^{-10} (a test of the formula independent of the propagator); the Gouy phase; the nonparaxial phase and fidelity of Eq. (30); and the Laguerre moments against their closed form.

3. Topology and angular momentum. The winding must equal l at all z, and the OAM must equal l and be conserved. Negative controls: a loop that misses the core must give zero, a Gaussian must give zero, and a loop through a dark ring of LG_1^1 (where samples sit on exact zeros) must be detected rather than give a wrong integer. A superposition must give the weighted mean OAM, and the lobe angle must follow $\arctan\zeta$.

4. Talbot effect and limits of the method. The exact z_T must agree with Eq. (33); a cosine grating must self-image at the exact z_T; the half-Talbot shift must be $d/2$; the binary grating must self-image only with the paraxial transfer function; the FFT must reproduce the Fourier series of the binary grating; the defect must scale as $\sqrt{\lambda/d}$; and the exact z_T formula must be numerically stable for large periods (it is written to avoid cancellation). For the sampling limits: the band-limit formula and z_c; the reference window must be alias-free; the band limit must be necessary beyond z_c and irrelevant below; padding must be necessary for large-angle wrap-around; and too small a window or too coarse a grid must be detected.

5 Results

5.1 Gaussian beam and the Gouy phase

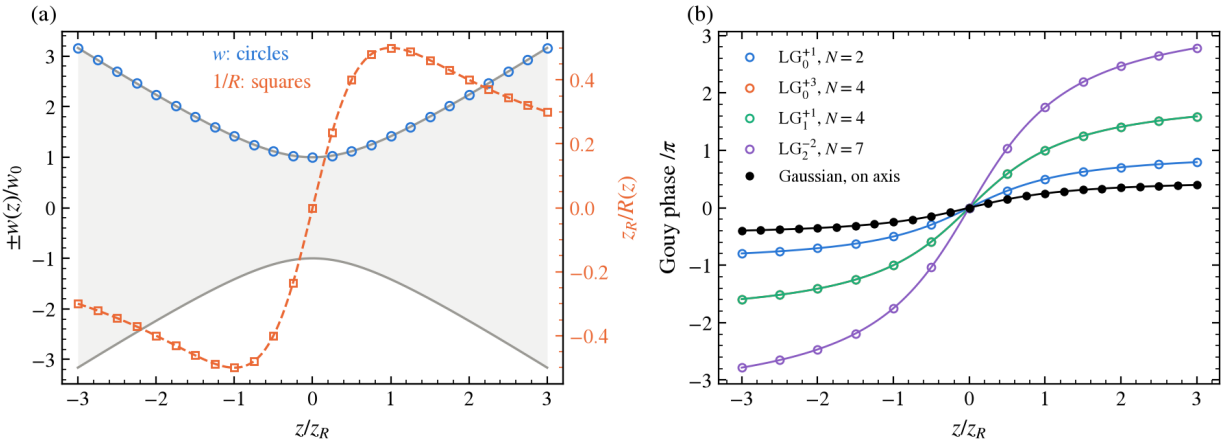


Figure 1: (a) Second-moment width (circles) and curvature z_R/R (squares) vs paraxial formulas. (b) Gouy phase: on axis for the Gaussian (filled), from overlaps for four LG modes (open), vs $(2p + |l| + 1)\arctan\zeta$.

Quantity, range	max deviation from paraxial
width, $0 \dots 3z_R$	1.14×10^{-4}
$1/R$, $\pm(0.25 \dots 3)z_R$	1.57×10^{-4}
Gouy, Gaussian, $\pm 3z_R$	3.16×10^{-5} rad
Gouy, LG_0^{+1}, LG_0^{+3}, LG_1^{+1}, LG_2^{-2}	2.9, 9.5, 11.4, 34.2×10^{-4} rad

5.2 What the paraxial formulas miss

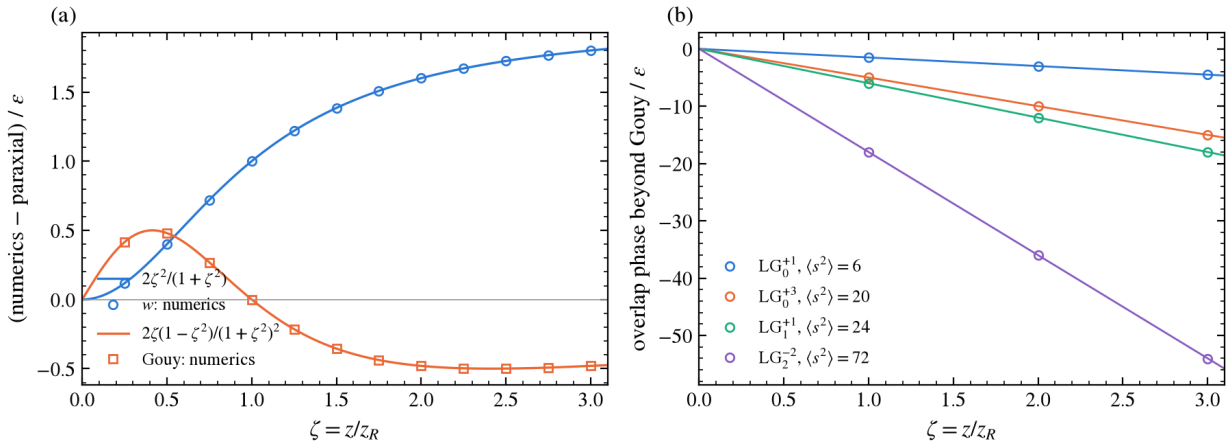


Figure 2: (a) Width and on-axis Gouy phase minus paraxial values, in units of ε (symbols), vs Eqs. (19)–(24) (lines, no fit). (b) LG overlap phase beyond the Gouy lag vs $-\zeta\langle s^2\rangle/4$.

mode	$\langle s^2\rangle$	phase at $3z_R$ (num / pred)	$1-F$ at $3z_R$ (num / pred)
LG_0^{+1}	6	$-2.8504/-2.8497\times10^{-4}$	$1.8964/1.8948\times10^{-7}$
LG_0^{+3}	20	$-9.5025/-9.4989\times10^{-4}$	$9.9375/9.9251\times10^{-7}$
LG_1^{+1}	24	$-1.1404/-1.1399\times10^{-3}$	$1.6804/1.6782\times10^{-6}$
LG_2^{-2}	72	$-3.4221/-3.4196\times10^{-3}$	$1.2150/1.2127\times10^{-5}$

The deviations from the paraxial formulas are not numerical error but nonparaxial physics with closed forms, which the numerics match to 2×10^{-11} (width), 10^{-8} rad (Gouy) and 2.5×10^{-6} rad (LG phase). The residual LG phase is the next term, $-\zeta\varepsilon^2\langle s^3\rangle/4$.

5.3 Laguerre–Gauss modes

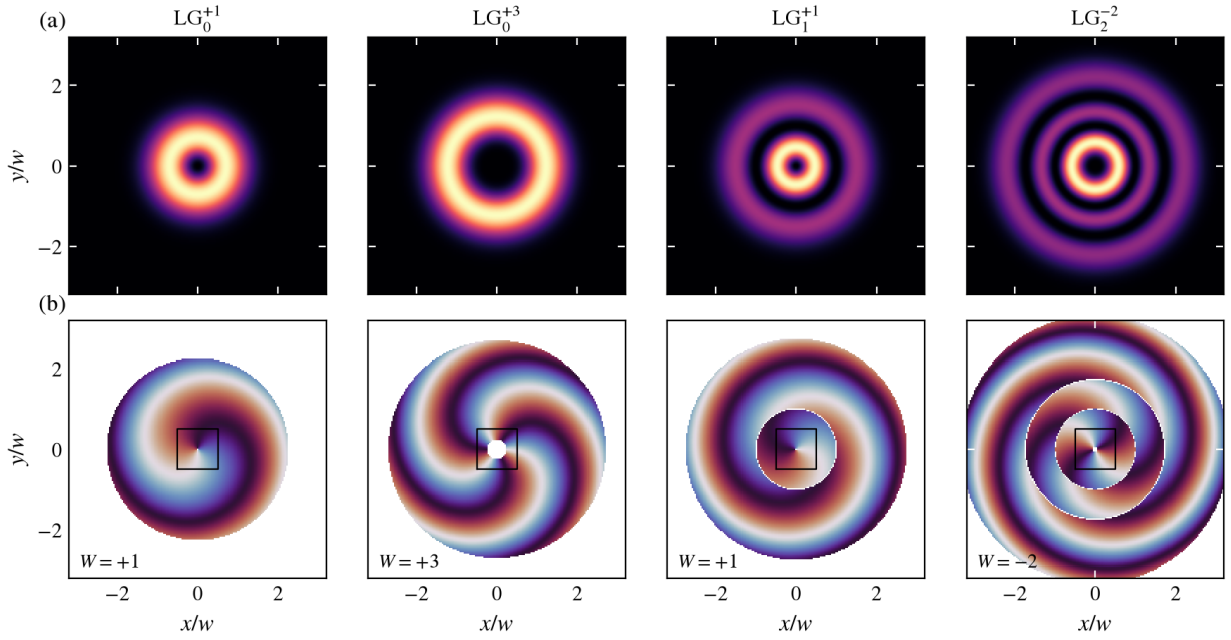


Figure 3: Four LG modes propagated to $z = z_R$. (a) Intensity. (b) Phase with carrier removed; the square is the winding loop.

mode	winding at $0, z_R, 3z_R$	$\langle L_z \rangle/\hbar$ at $0, 3z_R$
LG_0^{+1}	$+1, +1, +1$	$+1.000000000000$
LG_0^{+3}	$+3, +3, +3$	$+3.000000000000$
LG_1^{+1}	$+1, +1, +1$	$+1.000000000000$
LG_2^{-2}	$-2, -2, -2$	-2.000000000000

A loop through the dark ring of LG_1^1 (at $r = w$) was the first bug found: its samples sit on exact zeros. Loops wholly inside or outside the ring both give l. The OAM of $\sqrt{0.7}\,\mathrm{LG}_0^1 + \sqrt{0.3}\,\mathrm{LG}_0^{-2}$ is 0.100000000000 (a weighted mean), while any winding loop gives an integer.

5.4 Gouy phase made visible

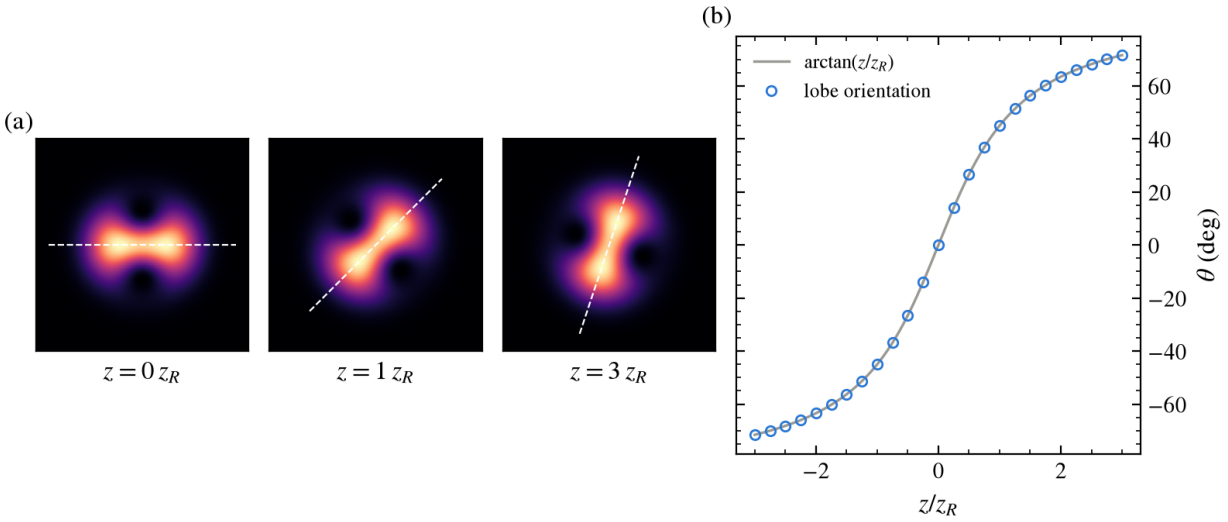


Figure 4: (a) $(\mathrm{LG}_0^0 + \mathrm{LG}_0^2)/\sqrt{2}$ at $z = 0, z_R, 3z_R$, scaled to $w(z)$; dashed lines at $\arctan\zeta$. (b) Lobe orientation vs $\arctan\zeta$.

The beam rotates 90° from waist to far field and 143.138° between $\pm 3z_R$. With the paraxial transfer function the angle equals $\arctan\zeta$ to 2×10^{-16}; with the exact one it deviates by at most 9.9×10^{-5} rad, an $O(\varepsilon)$ effect of the different nonparaxial phases of the two modes.

5.5 Talbot carpet

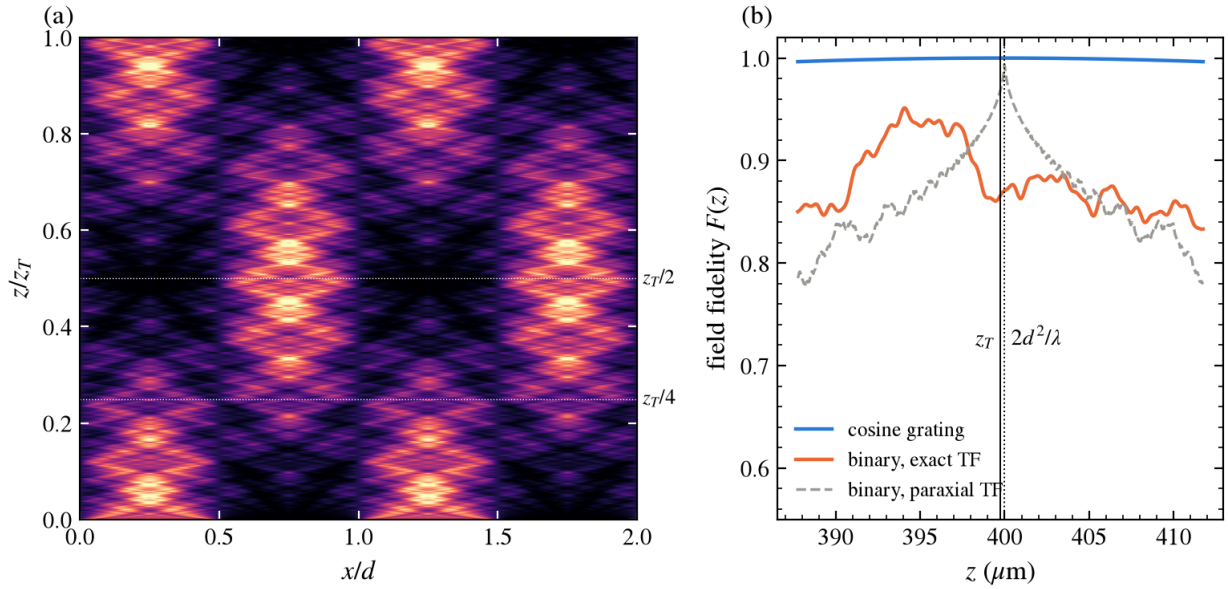


Figure 5: (a) Intensity behind a 50% binary grating ($d = 10\,\mu\text{m} = 20\lambda$), $0 \le z \le z_T$, exact transfer function. (b) Field fidelity near z_T: cosine, binary, binary with paraxial transfer function. Solid: exact z_T; dotted: $2d^2/\lambda$.

d	z_T exact	$2d^2/\lambda$	cosine revival peak	binary $F(z_T)$	binary, paraxial TF
$10\,\mu$m	399.74984	400	399.749846 ($\pm 2 \times 10^{-6}$)	0.866	$1 - O(10^{-15})$
$2\,\mu$m	15.74597	16	15.745862 ($\pm 1 \times 10^{-4}$)	0.819	$1 - O(10^{-15})$

The binary grating self-images neither at z_T nor nearby (best: $F = 0.951$ at $394.08\,\mu$m). Order m misses by $\frac{\pi}{2}m^2(m^2-1)(\lambda/d)^2$, already 2.4 rad for $m = 5$. The paraxial transfer function hides this because its phases are exactly $2\pi m^2$ at $2d^2/\lambda$. At $z_T/2$ the binary grating's fidelity with the shifted grating is only 0.914, although the first harmonic is shifted by exactly $d/2$.

5.6 Approach to perfect self-imaging

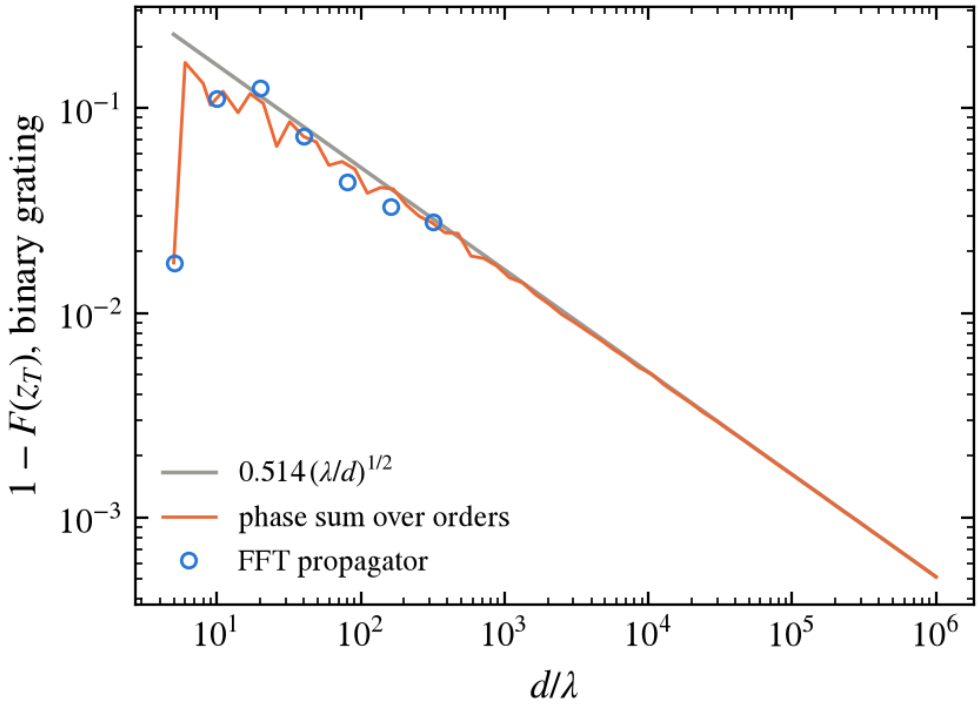


Figure 6: $1 - F(z_T)$ for a 50% binary grating vs d/λ: FFT propagator (circles), closed-form phase sum (line), asymptote Eq. (38) (grey).

d/λ	$1 - F(z_T)$	$0.5137\sqrt{\lambda/d}$	ratio
20	1.250×10^{-1}	1.149×10^{-1}	1.089
10^3	1.510×10^{-2}	1.624×10^{-2}	0.930
10^4	5.094×10^{-3}	5.137×10^{-3}	0.992
10^5	1.6244×10^{-3}	1.6244×10^{-3}	1.000
10^6	5.129×10^{-4}	5.137×10^{-4}	0.998

The fitted exponent for $d/\lambda \geq 10^3$ is -0.495. Even at $d = 100\lambda$, $1 - F = 0.045$.

5.7 Where the method fails

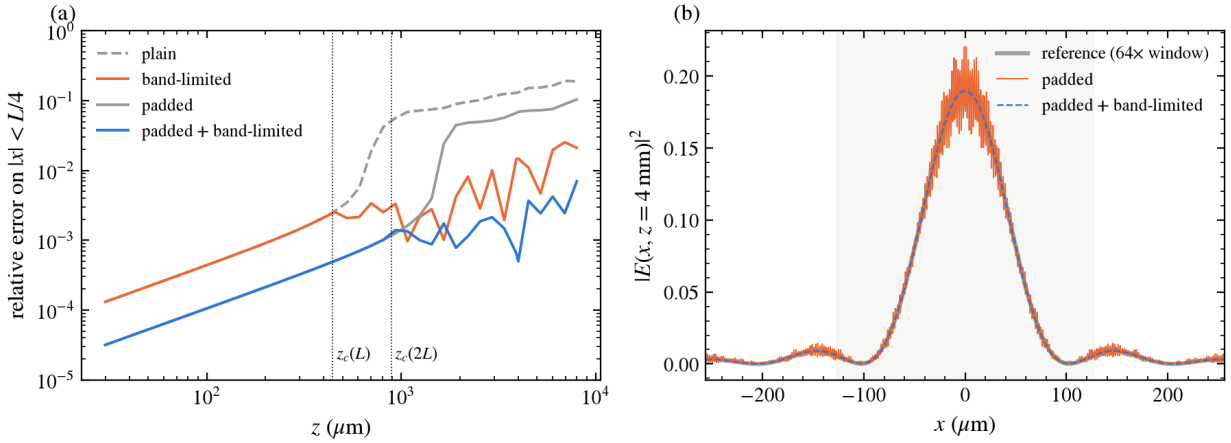


Figure 7: (a) Relative error of a 20 μm slit ($N = 1024$, $\Delta x = \lambda$, $L = 512\,\mu$m) vs a 64× wider reference; dotted: z_c for L and $2L$. (b) Intensity at $z = 4$ mm.

z (μm)	plain	band-limited	padded	padded + band-limited
300	1.4×10^{-3}	1.4×10^{-3}	3.2×10^{-4}	3.2×10^{-4}
4000	1.3×10^{-1}	1.5×10^{-2}	6.9×10^{-2}	5.0×10^{-4}
8000	1.9×10^{-1}	2.1×10^{-2}	1.0×10^{-1}	7.0×10^{-3}

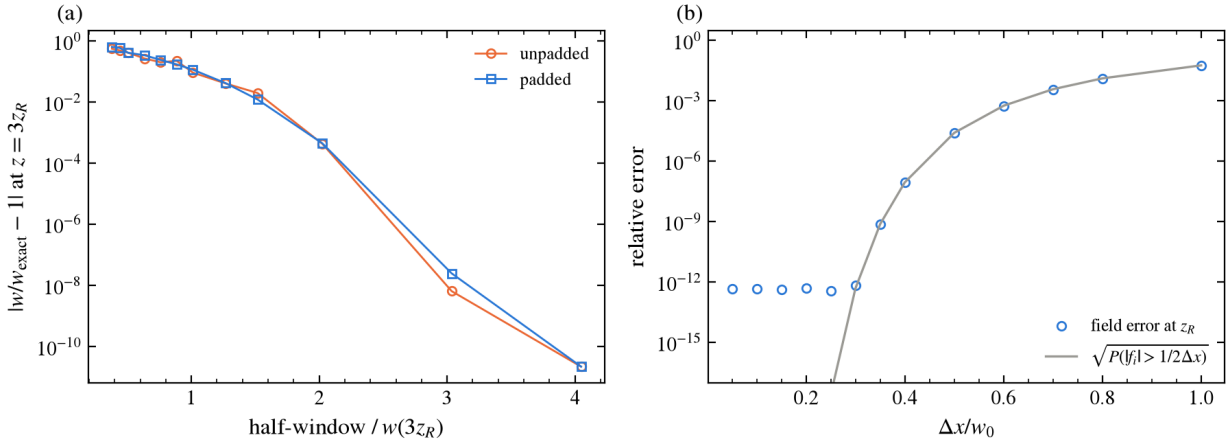


Figure 8: (a) Width error at $3z_R$ vs half-window/w. (b) Field error vs $\Delta x/w_0$, with $\sqrt{P_{\text{beyond Nyquist}}}$.

half-window/w	1.01	1.52	2.02	3.04	4.05
width error	9×10^{-2}	2×10^{-2}	4×10^{-4}	7×10^{-9}	2×10^{-11}
$\Delta x/w_0$	≤ 0.30	0.40	0.50	0.70	1.00
field error	$\leq 7 \times 10^{-13}$	8.8×10^{-8}	2.5×10^{-5}	3.7×10^{-3}	5.5×10^{-2}

In practice: hold $\pm 3w$ in the window at the largest z (for 10^{-8} accuracy), use $\Delta x \leq 0.3w_0$ (round-off level), and band-limit plus pad beyond z_c. Padding stops wrap-around but cannot recover light cropped by a too-small window.

6 Discussion

Why a focus adds a phase of π. A beam of width w has $k_\rho \sim 1/w$, so its k_z falls short of k by $\sim 1/(kw^2)$; integrating over z with w smallest at the focus gives $\int z_R\, dz/(z^2 + z_R^2) = \pi$. Mode order multiplies the spread in k_ρ^2 and hence the phase, which is what rotates the superposition.

Vortex versus Gaussian. A vortex has a dark core ($\propto r^{|l|}$), a topologically protected $2\pi l$ phase winding, $l\hbar$ OAM per photon conserved by propagation, and a Gouy phase growing $(|l|+1)$ times faster.

Exact method, finite sampling. Each plane wave propagates exactly, but the transfer function is sampled at $\Delta f = 1/L$ and its phase oscillates faster with z. Past z_c, light that should leave the window returns from the other side.

Limitations and outlook. The theory is scalar; for $w_0 \sim \lambda$ vector corrections (longitudinal field, spin–orbit coupling) compete with the ε terms, which motivates the vectorial focusing in the iSCAT project [9]. The coarse-grid error law is an empirical estimate (within 6%), not a derivation. Fractional Talbot images and astigmatic HG→LG mode converters [7] are natural extensions.

7 Conclusion

A carefully checked angular-spectrum propagator turns the gap between exact and paraxial optics into a set of closed-form, parameter-free corrections, all confirmed to near round-off. Vortex beams show integer winding and conserved OAM, and the Gouy phase becomes directly visible as a rigid rotation. The exact Talbot analysis overturns the textbook picture (only cosine gratings self-image exactly) and quantifies the binary-grating defect with a derived $\sqrt{\lambda/d}$ law. The sampling study gives practical rules for when the method can be trusted.

References

[1] J. W. Goodman, *Introduction to Fourier Optics*, 3rd ed., Roberts & Company (2005).

[2] A. E. Siegman, *Lasers*, University Science Books (1986).

[3] K. Matsushima, T. Shimobaba, *Band-limited angular spectrum method for numerical simulation of free-space propagation in far and near fields*, Opt. Express **17**, 19662 (2009).

[4] L. Allen, M. W. Beijersbergen, R. J. C. Spreeuw, J. P. Woerdman, *Orbital angular momentum of light and the transformation of Laguerre-Gaussian laser modes*, Phys. Rev. A **45**, 8185 (1992).

[5] M. V. Berry, S. Klein, *Integer, fractional and fractal Talbot effects*, J. Mod. Opt. **43**, 2139 (1996).

[6] D. G. Voelz, *Computational Fourier Optics: A MATLAB Tutorial*, SPIE Press (2011).

[7] M. W. Beijersbergen, L. Allen, H. E. L. O. van der Veen, J. P. Woerdman, *Astigmatic laser mode converters and transfer of orbital angular momentum*, Opt. Commun. **96**, 123 (1993).

[8] M. Lax, W. H. Louisell, W. B. McKnight, *From Maxwell to paraxial wave optics*, Phys. Rev. A **11**, 1365 (1975).

[9] L. Novotny, B. Hecht, *Principles of Nano-Optics*, 2nd ed., Cambridge University Press (2012).